# *W6 PRACTICE*

## *My SQL*

### *At the end of his practice, you should be able to…*

- ✔ Establish a **MySQL connection** on the back-end app
- ✔ Implement a **repository** layer using **MySQL queries**
- ✔ **Test the endpoints** (REST API client + front-end app)
- ✔ **Extends the project** to handle **4 tables** in the database

### *How to start?*

- ✔ Download **start code** from related MS Team assignment
- ✔ Run `npm install` on both front and back projects
- ✔ Run `npm run dev` on both front and back projects to run the client and the server

### *How to submit?*

- ✔ Submit your **code** on MS Team assignment

### *Are you lost?*

*To review MySQL queries syntax*
https://www.w3schools.com/mysql/mysql_sql.asp

*To install My SQL Server (if needed)*
https://dev.mysql.com/doc/refman/8.4/en/windows-installation.html
https://dev.mysql.com/downloads/

*To connect Node back end to MySQL*
https://www.w3schools.com/nodejs/nodejs_mysql.asp
https://sidorares.github.io/node-mysql2/docs/documentation

# *EXERCISE 1 – MySQL* ***Manipulation***

**Before starting !**

You should have a MySQL Server running. Check it out with bellow command:

```
mysql -u root -p
```

You should see MySQL monitor run properly:

```
C:\Users\PC>mysql -u root -p
Enter password: ***********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 9.3.0 MySQL Community Server - GPL
```

If not, you need to install and configure MySQL server properly.
[https://dev.mysql.com/doc/refman/8.4/en/windows-installation.html](https://dev.mysql.com/doc/refman/8.4/en/windows-installation.html)

### Q1 - Create the database and the table of articles

- Open the terminal and launch MySQL monitor:

```
mysql -u root -p
```

- Create a new database (e.g. **week6Db**) using the command line
- Create a new table (articles) with the columns below:

```
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| id         | int          | NO   | PRI | NULL    | auto_increment |
| title      | varchar(255) | YES  |     | NULL    |                |
| content    | text         | YES  |     | NULL    |                |
| journalist | varchar(100) | YES  |     | NULL    |                |
| category   | varchar(50)  | YES  |     | NULL    |                |
+------------+--------------+------+-----+---------+----------------+
```

### Q2 - Review My SQL queries

- Complete the bellow table with the appropriate MySQL query

| Use case | My SQL Query |
|---|---|
| Get all articles | SELECT * FROM articles |
| Get articles written by the journalist ‘RONAN” | Select * from aritbles where journalist = ‘RONAN’ |
| Add an article | Insert into articles (title,contnet, journalist,category) |
| Delete all articles whose title starts with “R” | Delete from articles where title like “R%” |

# *EXERCISE 2 – MySQL on **Backend***

*For this exercise, you start with a start frontend and a backend code.*
**The goal for this exercise is to replace the provided mock repository with a MySQL repository.**

### Q1 - Run Frontend & Backend

Open a dedicated terminal to run the server:

```
cd back
npm i
npm run dev
```

Open a dedicated terminal to run the client:

```
cd front
npm i
npm run dev
```

Open the browser and check the front end is correctly **connected with the back end** :


The project already works as we provide fake data (mock repository).
*Let's understand in detail the back and front ends.*

*FRONT-END*

**Q2 - Look at ArticleForm**

How does the component know whether to create a new article or update an existing one?

```
By looking at the isEdit props value passing from App. If it a truthy value
aritcleForm will call updateArticle function, otherwise it will call
createArticle function.
```

Why is the **useParams hook** used in this component? What value does it provide when isEdit is true?

```
When isEdit is true, it gives the id to updateArticle function
```

Explain what happens inside the **useEffect** hook. When does it run, and what is its purpose?

```
The useEffect hook manages data synchronization by fetching existing article
details when the component is in edit mode. It runs upon the component's
initial mount and re-runs whenever the id (from useParams) changes, ensuring
the form is always populated with the correct data for the specific article
being edited.
```

**Q3 - Look at the ArticleList**

How are the three promise states (loading, success, and error) handled in the fetchArticles function?

```
First setIsLoading to true, and set erro to empty, after trying to load data
from getAriticle, if it succeeds, fetch that data to setArticles, if it fails
setError to a warning text.
```

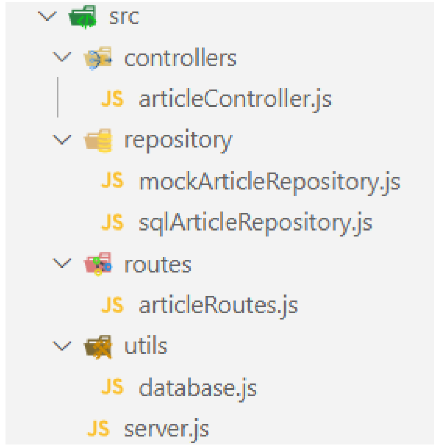
What is the role of the ArticleCard component, how does it communicate with the parent ArticleList?

```
To display articleCard with article's object detail. Articlelist called
articlecard and passing value and function to  article, onView, onEdit,
onDelete props.
```

*BACK-END*

**Q4 - Why 3 layers ?**

The backend is composed of the below 3 layers: routes, controllers and repository :




Describe the **responsibility** of each **layer** by completing the table below:

| LAYTER | RESPONSABILITIES |
|---|---|
| Routes | Use to manage the route endpoint |
| Controller | To manage the logic for each corresponding route endpoint |
| Repository | Manage logic that relates to data |

**Q5 - Implement the database connection**

Here are the files you need to update to **connect the backend to the database**:

| FILE | RESPONSABILITIES |
|---|---|
| /.env | securely store your MySQL database credentials |
| | |
| /utils/ database.js | Holds the **MySQL connection setup** logic<br><br>*Responsible for creating and exporting a **connection pool** that other parts of the application can use.* |
| /repository/sqlArticleRepository.js | Provides **a clean, reusable interface** to interact with the articles table in your **MySQL database**. |

| | |
|---|---|
| | *Encapsulates all the SQL queries related to articles and exposes them as functions that the rest of your application can call.* |

<u>Here is what you need to do:</u>

- **.env file**
  Create a .env file to securely store your MySQL database credentials.
  *See [https://www.npmjs.com/package/dotenv](https://www.npmjs.com/package/dotenv)*

```
DB_HOST=localhost
DB_USER=root
DB_PASSWORD=complete this line
DB_NAME=complete this line
PORT=4000
```

- **utils/database.js**
  - Create  a **MySQL connection pool** using the credentials from the .env file.
  [https://sidorares.github.io/node-mysql2/docs#using-connection-pools](https://sidorares.github.io/node-mysql2/docs#using-connection-pools)
  - Export this connection pool so it can be used by other modules in the project.

- **repositories/sqlArticleRepository.js**

  Implement the following functions to interact with the articles table in the database:

```
getAll() – fetch all articles
getById(id) – fetch one article by ID
create(article) – insert a new article
update(id, article) – update an existing article
remove(id) – delete an article by ID
```

  Use the connection pool from database.js to **execute the SQL queries** inside these functions.

  *As an example, to implement getAll()  :*

```
export async function getArticles() {
  const [rows] = await pool.query("SELECT * FROM articles");
  return rows;
}
```

**Q6 -  Test the endpoints**

To test the implementation of MySQL repository (*create, update, remove, get articles…)*

- First, perform tests using a **REST API client** (thunder or postman)

- Then, run the **front-end project** and asset the views work properly

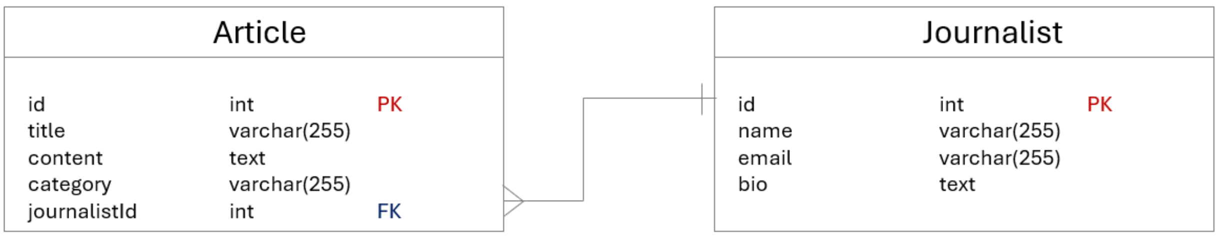
# *EXERCISE 3 – Handle **Journalists***

*For this exercise, you continue on the previous exercise code.*

Now, users want to see **who wrote each article** to better understand the source.

- You will need to update the app, so the article page shows the **journalist's name and info.**
- You will need to provide a journalist view, showing all articles written by a specific journalist.

**Database**




*Update your database structure to handle the journalist table*

- Create **journalists table** with fields: id, name, email, bi
- Update **articles table** to include journalist_id foreign key
- **Populate the database** with some fake data

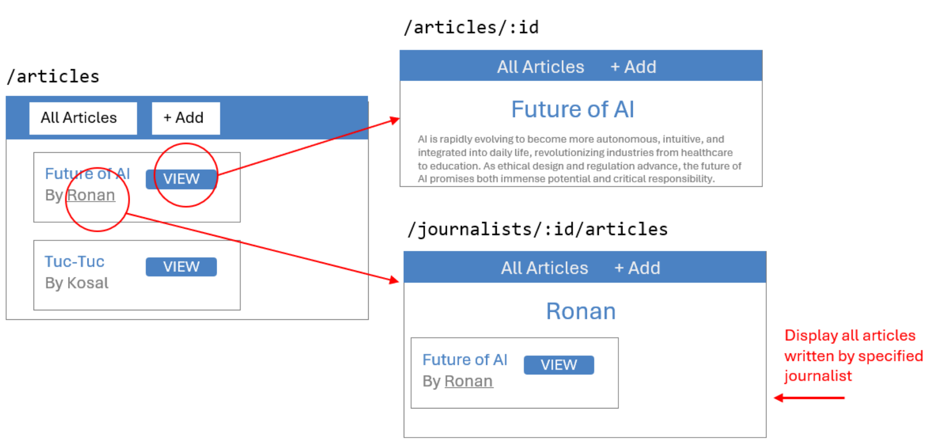
**Back End**

- Implement **repository** methods:
  - Fetch articles with joined journalist name (using **SQL JOIN**)
  - Fetch all articles written by a specific journalist name (using **SQL JOIN**)

- Add **controller** functions:
  - Get all articles by journalist ID

- Define **new API routes**:
  - `GET /api/articles/:id` article + journalist name.
  - `GET /api/journalists/:id/articles` articles list by journalist.

**Front End**



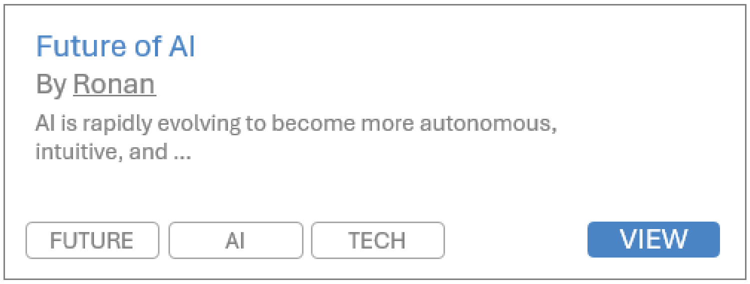

*An additional view shall display all articles written by the specific journalist*

- Update **Article Details page**:
    - Display journalist name alongside the article.
- Create **Journalist Articles List page**:
    - Display all articles by selected journalist.
- Add navigation:
    - From Article Details page, allow users to click journalist name to view **that journalist's articles.**
- Update API calls:
    - Fetch combined article + journalist data.
    - Fetch articles filtered by journalist ID.

# *EXERCISE 4 – Handle **Tags*** **BONUS**

*For this exercise, you continue on the previous exercise code.*

Now, users want to easily **assign tags to articles**.
The users can then filter articles  by selecting different tags.




You will need to add categories to articles and let users filter the article list by selecting a category.

Database

- Create a new **table** Category  (id, name).
  - *What kind of relationships do we have between articles and categories?*

**Back End:**

- Implement repository methods to:
  - Retrieve all categories.
  - Retrieve all articles filtered by category, using JOIN to include category name.
- Add a new API endpoint to get articles by category ID.

**Front End:**

- Add a **multiple categories filter UI component** on the **article list page** (multiple choice dropdown, chipset selector).
- When categories are selected, fetch and display only articles in those categories.
- Display categories names alongside articles in the list.